

MSCIT sem 3
Employment Management System Project REPORT
Mobile and web development
ID:202412061
Name: Parmar Shenal Kamlesh

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Employee Leave Management System (MERN Stack)

GitHub Repository

👉 **Project Link:** *GitHub repository URL here*

https://github.com/shenal-parmar/Employee_Management_System

Clone link : `git clone`

https://github.com/shenal-parmar/Employee_Management_System.git

Deployment link: <https://employee-management-system-zeta-nine.vercel.app>

Project Overview

The **Employee Leave Management System** is a full-stack MERN application that allows employees to register, apply for leaves, and track their leave status. The admin can approve/reject registrations and leaves. The system supports **real-time notifications** using Socket.IO and **email notifications** for important events.

This project demonstrates how to integrate authentication, REST APIs, database operations, email automation, and real-time updates in a complete industry-level MERN stack application.

Main Functionalities

1. Employee Registration

- Employees create an account by entering basic details.
 - Their status remains **Pending** until approved by the Admin.
 - Email is sent to Admin notifying new registration.
 - Socket.IO notifies Admin in real time that a new user has registered.
-

2. Admin Panel

Admin can:

- View all pending employees
- Approve or reject employee registration
- View list of all employees
- Receive real-time notifications when employees apply for leave

Once an employee is approved:

- Employee receives an **email notification** stating approval.
-

3. Employee Login & Authentication

- JWT-based login system
 - Role-based authorization ensures **Admin** and **Employee** have different privileges
 - Passwords hashed with bcrypt for security
-

4. Leave Request Management

Employees can:

- Apply for leave
- Select leave type, dates, and description
- View their leave applications and statuses

Admin can:

- View all leave requests
- Approve / reject with a single click

Upon approval:

- Employee receives an **email notification**
- Employee receives **real-time notification** via Socket.IO

5. Real-Time Notifications (Socket.IO)

The system uses Socket.IO to:

- Notify Admin instantly when a new employee registers
- Notify Employee instantly when their leave is approved

6. Email Notifications

Emails sent for:

- New employee registration (sent to Admin)
- Employee approval (sent to the employee)
- Leave approval (sent to the employee)
- Admin received notification when employee applies for leave

Email service uses:

- Nodemailer with Gmail SMTP (or any provider you set)
-

7. Secure REST APIs

Backend built using Express:

- Authentication routes
 - Employee routes
 - Leave routes
 - Protected routes with JWT middleware
-

8. Clean UI & Dashboard

Frontend built using:

- React.js
- Axios
- Tailwind CSS / Bootstrap
- React Query for API caching

Employees and Admin see separate dashboards.

Tech Stack

Frontend

- React.js
- React Router
- Axios

- React Query
- Tailwind CSS / Bootstrap

Backend

- Node.js
- Express.js
- MongoDB + Mongoose
- JWT Authentication
- Socket.IO
- Nodemailer

Third-Party Libraries / Tools Used

Backend

Library / Tool	Purpose
express	Create REST APIs
mongoose	MongoDB ORM
dotenv	Environment variables
bcrypt	Password hashing
jsonwebtoken	JWT auth
cors	Enable cross-origin requests
cookie-parser	Parse cookies
nodemailer	Sending emails
socket.io	Real-time communication
express.json / express.urlencoded	Body parsers

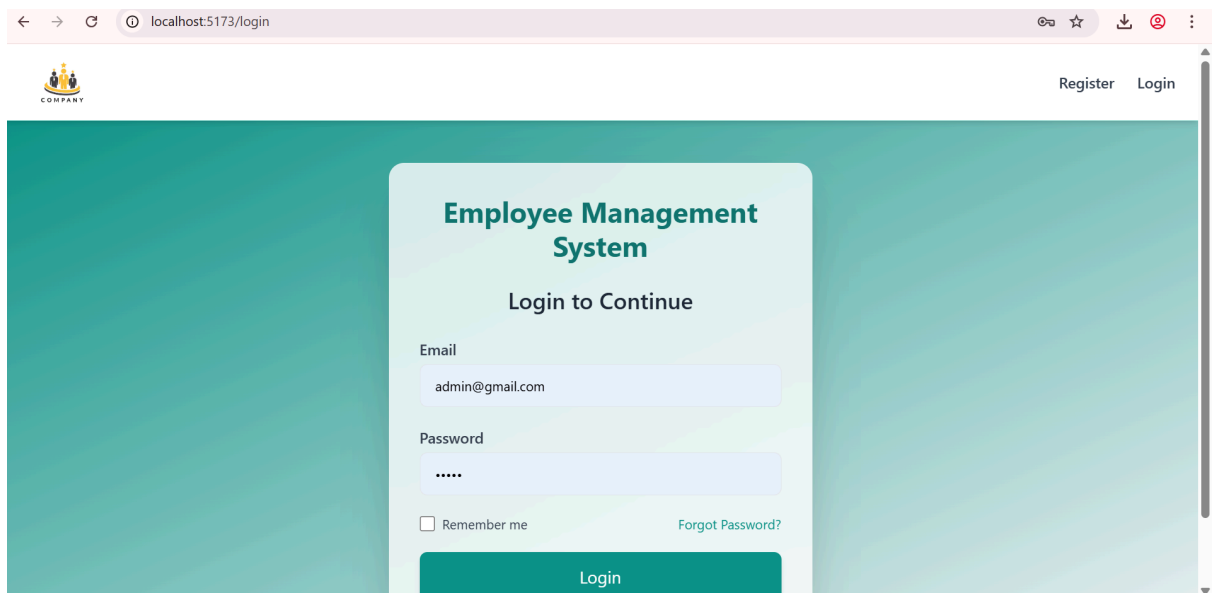
express.static

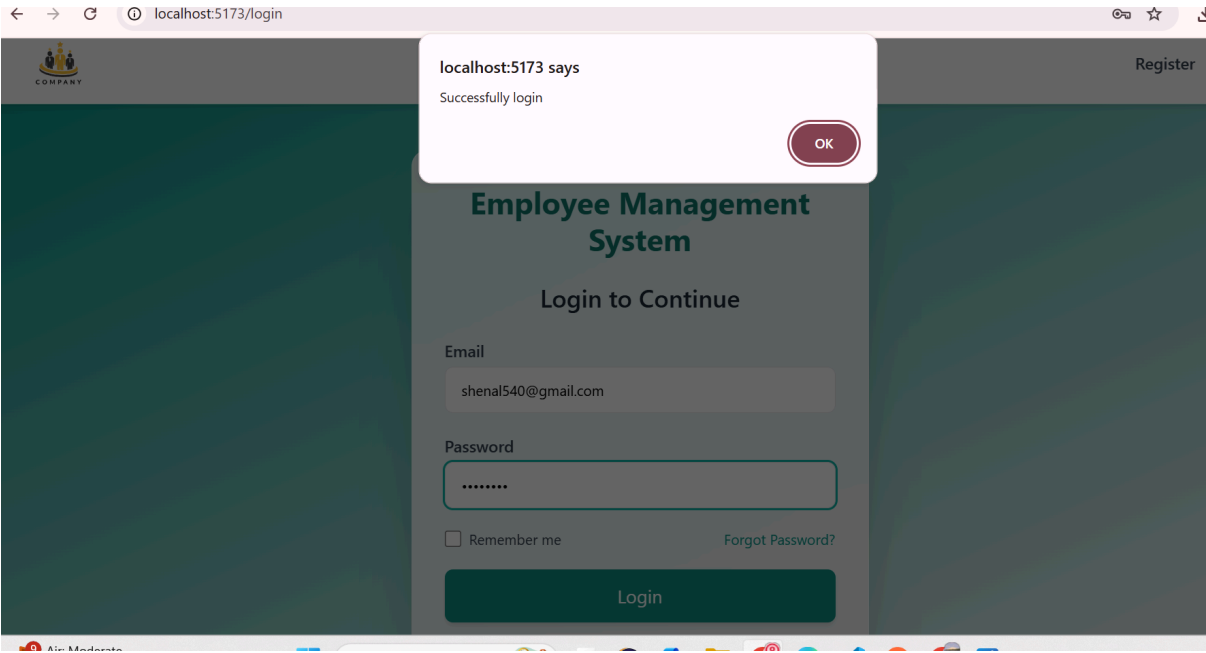
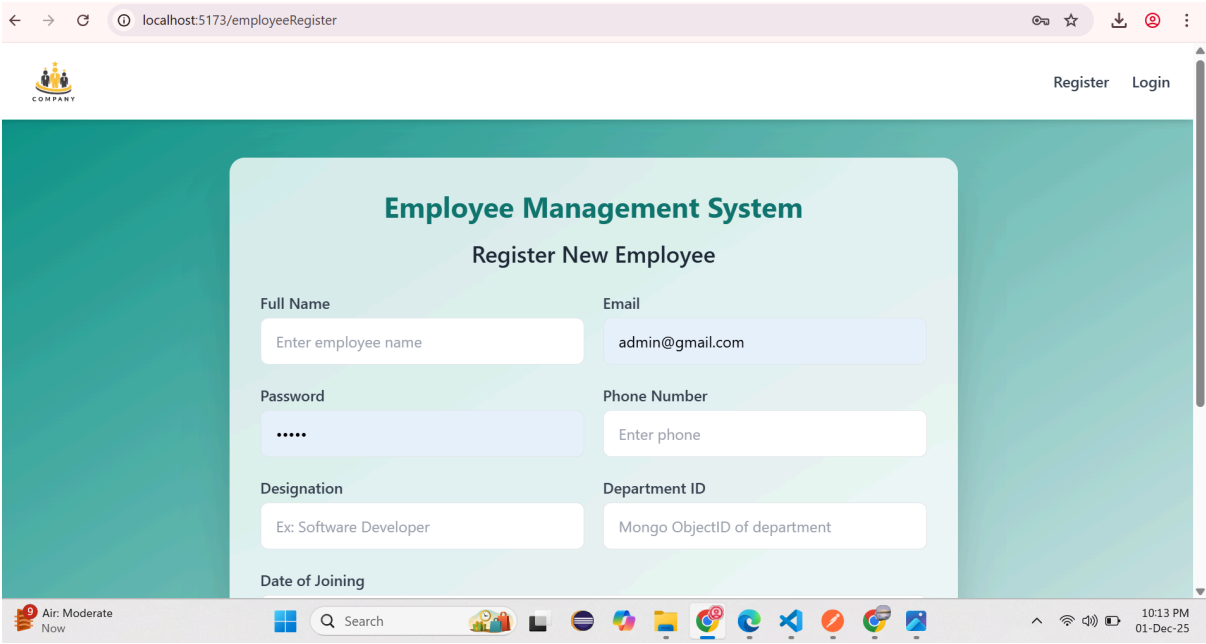
Serving static files

Frontend

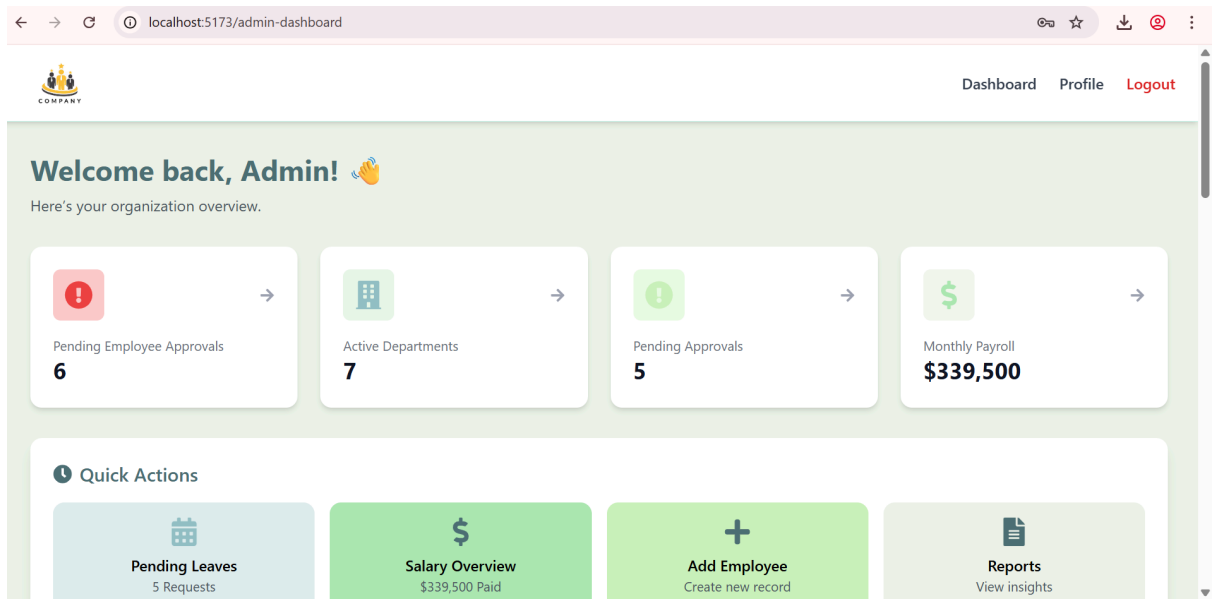
Library / Tool	Purpose
react	UI framework
react-router-dom	Routing
axios	HTTP client
react-query	Query caching and API state
tailwindcss	Styling
socket.io-client	Real-time connection

- Screenshots :





Admin dashboard



This screenshot shows the top section of an admin dashboard. At the top, there's a navigation bar with a logo on the left and links for 'Dashboard', 'Profile', and 'Logout' on the right. Below the navigation bar, a large green banner contains a welcome message 'Welcome back, Admin!' with a hand icon and a sub-header 'Here's your organization overview.' Underneath the banner, there are four white cards with green borders, each containing an icon, a title, and a value. The first card shows 'Pending Employee Approvals' with a value of 6. The second card shows 'Active Departments' with a value of 7. The third card shows 'Pending Approvals' with a value of 5. The fourth card shows 'Monthly Payroll' with a value of \$339,500. Below these cards, there's a 'Quick Actions' section with four colored buttons: 'Pending Leaves' (5 Requests), 'Salary Overview' (\$339,500 Paid), 'Add Employee' (Create new record), and 'Reports' (View insights).

localhost:5173/admin-dashboard

Dashboard Profile Logout

Welcome back, Admin! 🙌
Here's your organization overview.

Pending Employee Approvals
6

Active Departments
7

Pending Approvals
5

Monthly Payroll
\$339,500

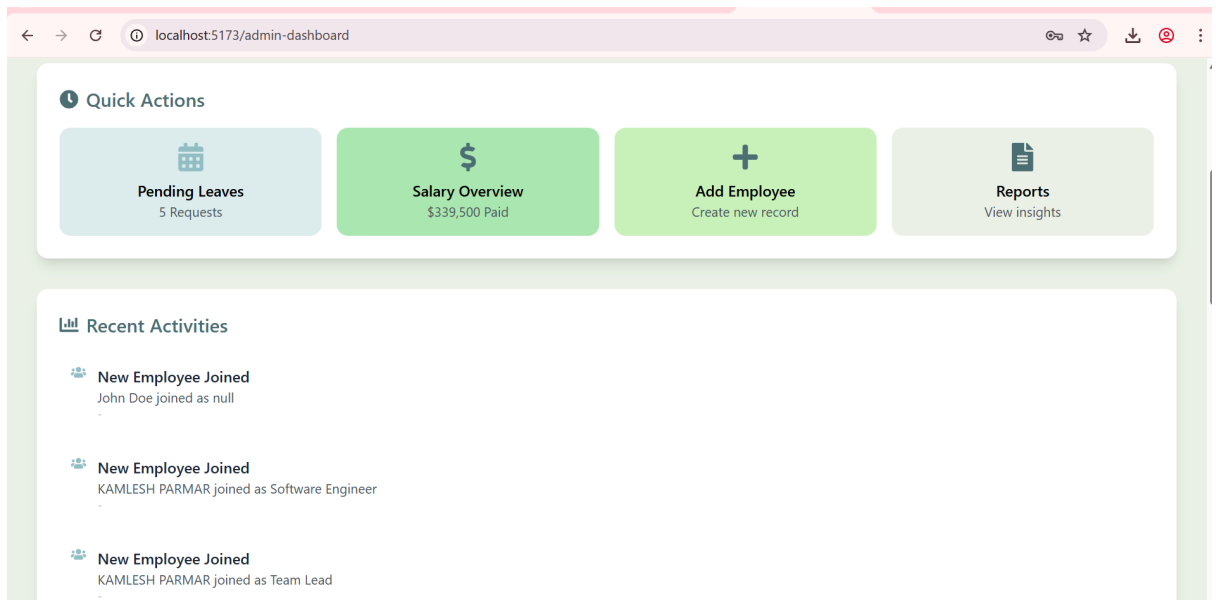
Quick Actions

Pending Leaves
5 Requests

Salary Overview
\$339,500 Paid

Add Employee
Create new record

Reports
View insights



This screenshot shows the bottom section of the admin dashboard. It features a 'Recent Activities' section with a list of three entries, each starting with a person icon and the title 'New Employee Joined'. The first entry is 'John Doe joined as null'. The second entry is 'KAMLESH PARMAR joined as Software Engineer'. The third entry is 'KAMLESH PARMAR joined as Team Lead'. Above this section, there's a 'Quick Actions' section with four colored buttons: 'Pending Leaves' (5 Requests), 'Salary Overview' (\$339,500 Paid), 'Add Employee' (Create new record), and 'Reports' (View insights).

localhost:5173/admin-dashboard

Quick Actions

Pending Leaves
5 Requests

Salary Overview
\$339,500 Paid

Add Employee
Create new record

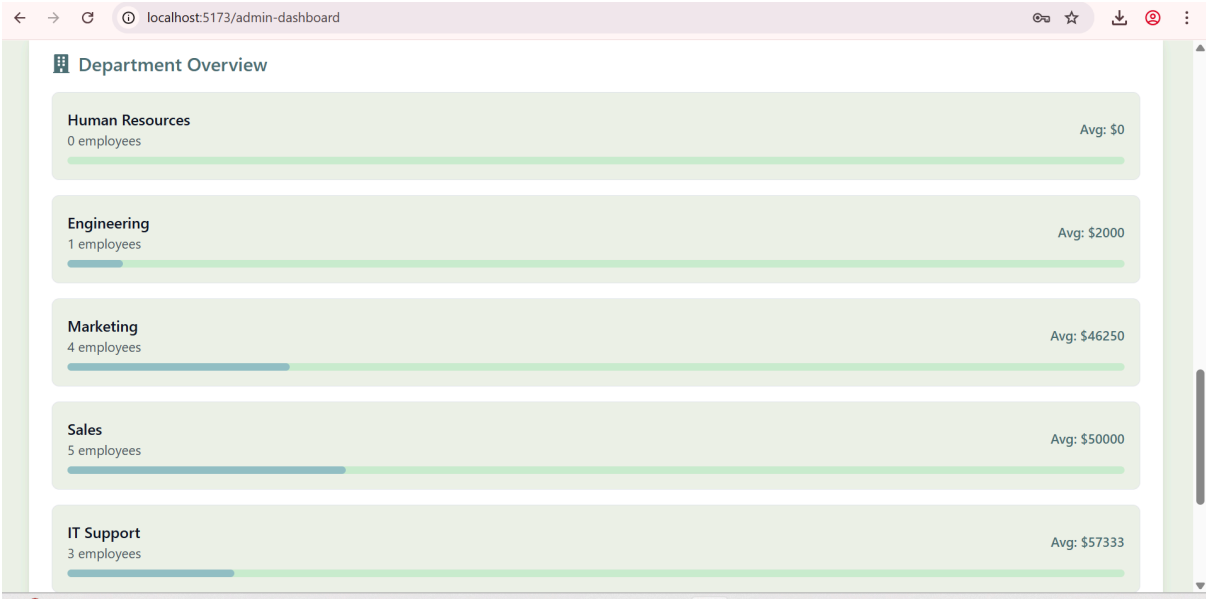
Reports
View insights

Recent Activities

New Employee Joined
John Doe joined as null

New Employee Joined
KAMLESH PARMAR joined as Software Engineer

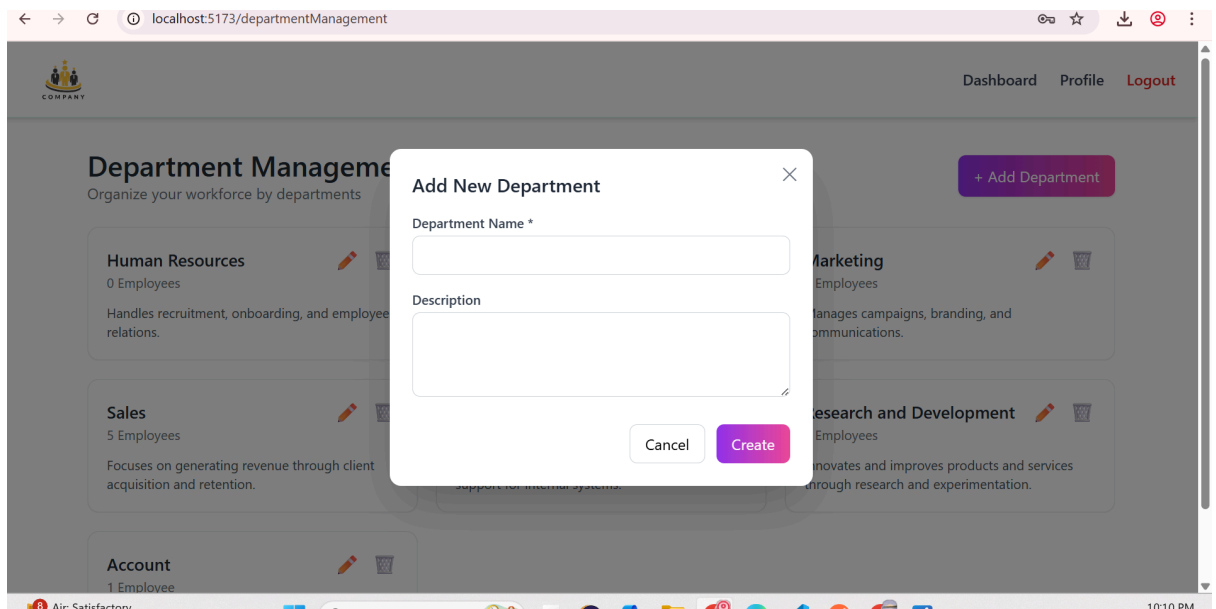
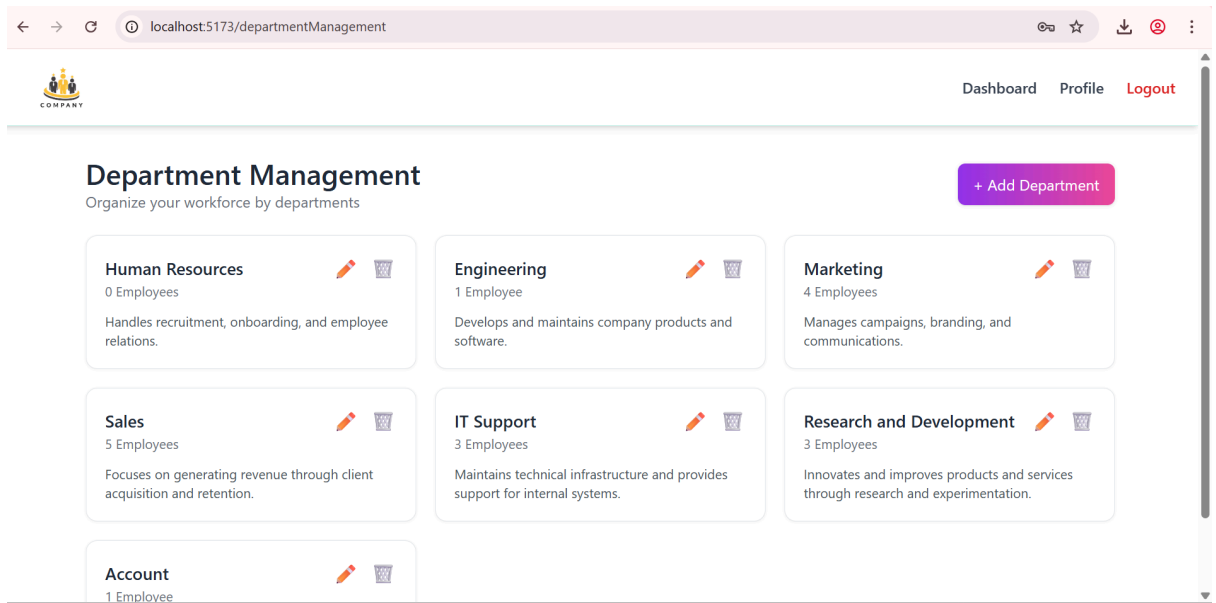
New Employee Joined
KAMLESH PARMAR joined as Team Lead



Pending approval employees:

The screenshot displays a web browser window with the address bar showing 'localhost:5173/pending-employees'. The page has a header with a logo and navigation links: 'Dashboard', 'Profile', and 'Logout'. The main content area is titled 'Pending Employee Approvals' and contains a table with the following data:

Name	Email	Phone	Action
Meera Jain	meera.jain@company.com	9012345678	Approve
Karan Patel	karan.patel@company.com	9123456789	Approve
chitan	chitan@gmail.com		Approve
Mehul Soni	mehul.soni@company.com	9090909090	Approve
KAMLESH PARMAR	shenal540@gmail.com	09904707943	Approve
John Doe	john.doe@example.com	9876543210	Approve



← → ↺

localhost:5173/leaves

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COMPANY

Dashboard Profile Logout

Leave Management

Review and manage employee leave requests

+ Add Leave

Pending Requests5

Approved5

Rejected11

All Requests22

🔍 Search by employee name, ID, or leave type...

Pending (5) Approved (5) Rejected (11) All (22)

Employee	Leave Type	Duration	Days	Reason	Status	Actions
Krishna Parmar	Casual Leave	11/24/2025 - 11/26/2025	3	trip	Pending ▾	Edit
Krishna Parmar	Casual Leave	11/24/2025 - 11/26/2025	3	trip	Pending ▾	Edit
Krishna Parmar	Casual Leave	11/29/2025 - 11/29/2025	1	health	Pending ▾	Edit

🔄 ⓘ localhost:5173/leaves

Dashboa

Leave Management

and manage employee leave requests

ing Requests

App5

Search by employee name, ID, or leave type.

Pending (5) Approved (5) Rejected (

oyee

Leave Type

ia Parmar

Casual Leave

ia Parmar

Casual Leave

ia Parmar

Casual Leave

11/24/2025 - 11/26/2025

4

health

Pending ▾

Add Leave

Select Employee ▾

Employee ID

Select Leave Type ▾

dd----yyyy 📅 dd----yyyy 📅

0

Reason

Pending ▾

Cancel Add Leave

All Requests22

Status

Pending ▾

Pending ▾

Pending ▾

← → ↺

localhost:5173/salaryManagement

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COMPANY

Dashboard Profile Logout

Salary Management

Manage employee salaries and payments

\$

Total Paid

\$339,500

📄

Total Records

9

📄

Pending

2

🔍

Search by employee name or month...

Employee	Month/Year	Basic	Allowance	Deduction	Net Salary	Status	Actions
Aarav Shah	February 2024	\$45000	+\$5000	-\$2000	\$48000	Paid	✎
Aarav Shah	March 2024	\$45000	+\$4500	-\$1500	\$48000	Paid	✎
Riya Mehta	February 2024	\$68000	+\$8000	-\$3000	\$73000	Paid	✎
Riya Mehta	March 2024	\$68000	+\$8500	-\$2500	\$74000	Paid	✎

← → ↺

localhost:5173/salaryManagement

🔍 ⭐ 📄 🗑

COMPANY

Dashboard Profile Logout

Salary Management

Manage employee salaries and payments

\$

Total Paid

\$339,500

📄

Total Records

9

📄

Pending

2

🔍

Search by employee name or month...

Employee	Month/Year	Basic	Allowance	Deduction	Net Salary	Status	Actions
Aarav Shah	February 2024	\$45000	+\$5000	-\$2000	\$48000	Paid	✎
Aarav Shah	March 2024	\$45000	+\$4500	-\$1500	\$48000	Paid	✎
Riya Mehta	February 2024	\$68000	+\$8000	-\$3000	\$73000	Paid	✎
Riya Mehta	March 2024	\$68000	+\$8500	-\$2500	\$74000	Paid	✎

Edit Salary Record

Select Employee

Aarav Shah ()

45000

5000

2000

February

2024

Paid

Net Salary: \$48,000

Cancel Update

employee dashboard:

← → ↺ 🌐 localhost:5173/employee-dashboard 🔍 ⭐ ⬇️ 🛑 ⋮

Dashboard Profile Logout

Welcome, **KAMLESH PARMAR** 🙌
Here's an overview of your work, leaves, and salary.


Pending Leaves
0


Approved Leaves
1


Total Salary Received
\$0

My Leaves Apply for Leave

TYPE	FROM	TO	STATUS
Paid Leave	Nov 30, 2025	Dec 02, 2025	Rejected
Casual Leave	Nov 30, 2025	Dec 03, 2025	Rejected

Profile page 👍

Dashboard Profile Logout


KAMLESH PARMAR

 shenal540@gmail.com  09904707943

 691210ee6cd4140df5d28930

Reports

← → ↺ 🌐 localhost:5173/reports 🔍 ⭐ ⬇️ 🛑 ⋮

Dashboard Profile Logout

Reports & Analytics
Download detailed reports in CSV format


19
Total Employees


7
Departments

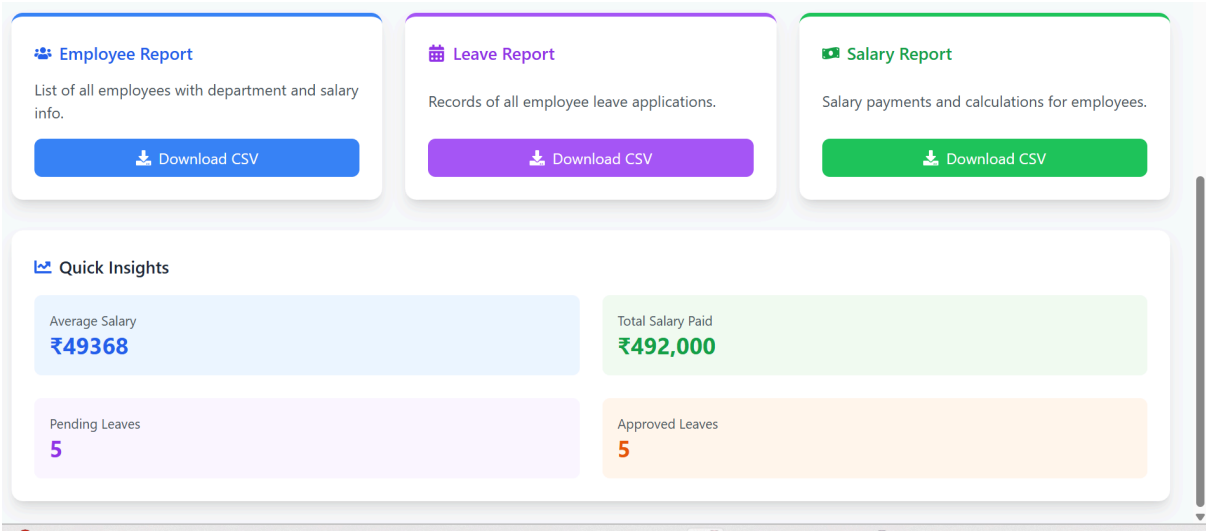

22
Total Leaves


9
Salary Records

 **Employee Report**
List of all employees with department and salary info.
Download CSV

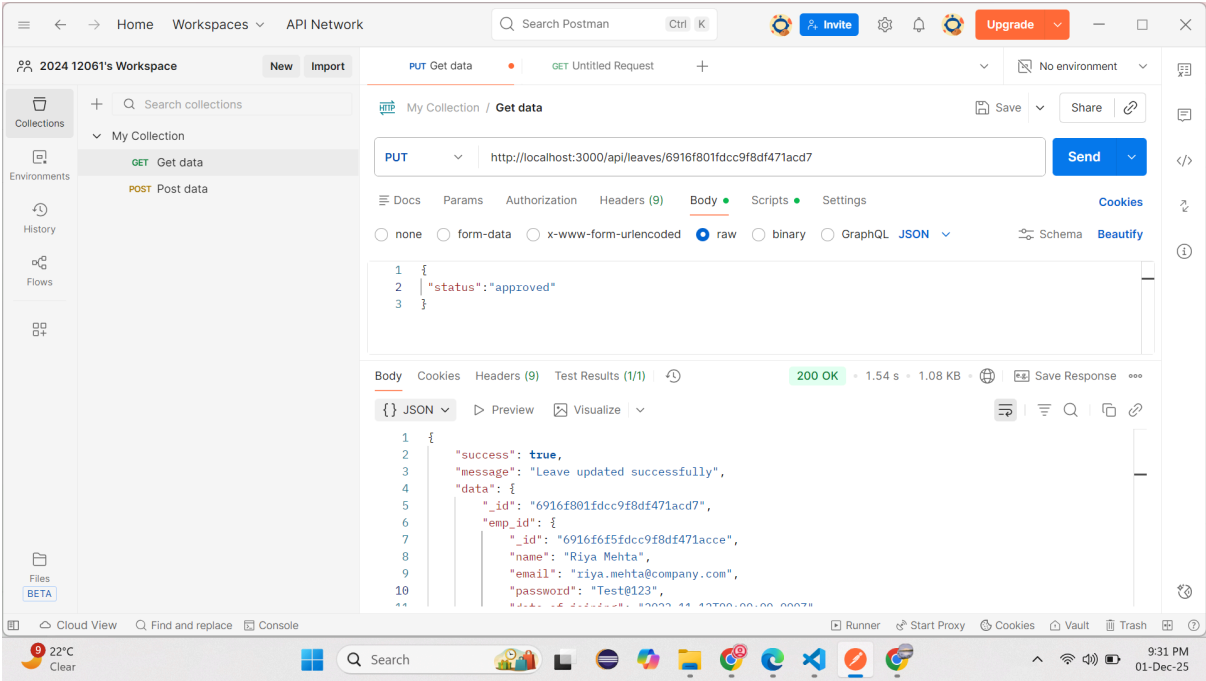
 **Leave Report**
Records of all employee leave applications.
Download CSV

 **Salary Report**
Salary payments and calculations for employees.
Download CSV



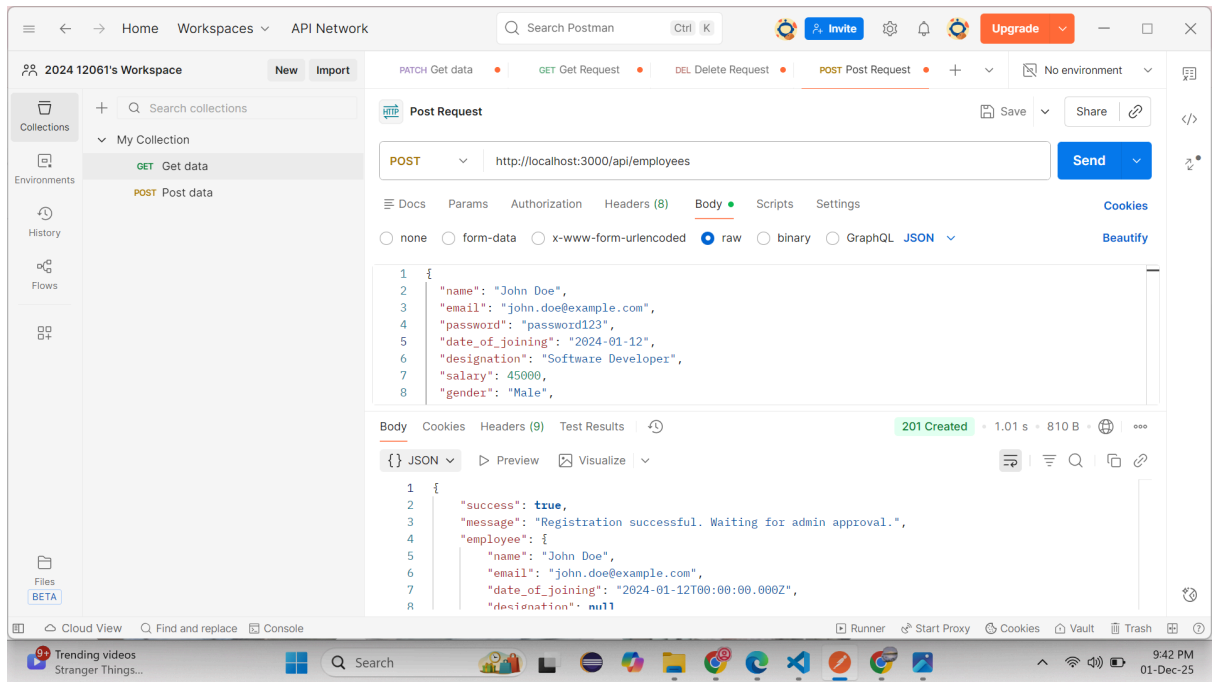
Postman screenshots;

PUT:

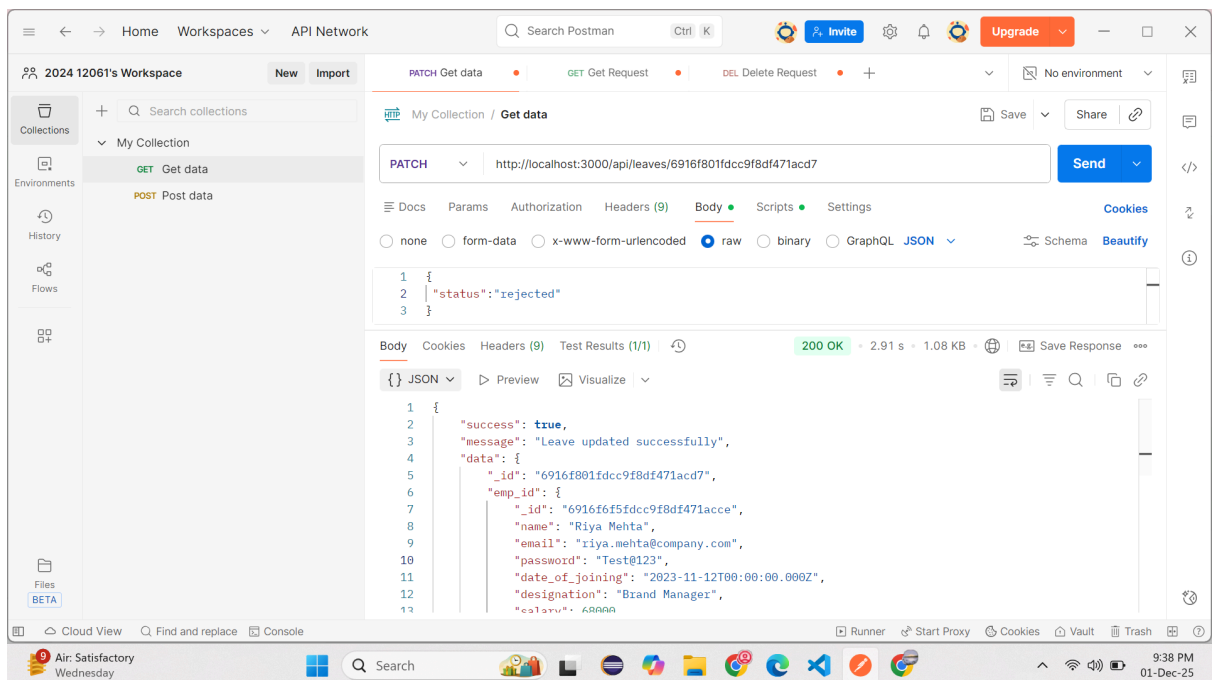


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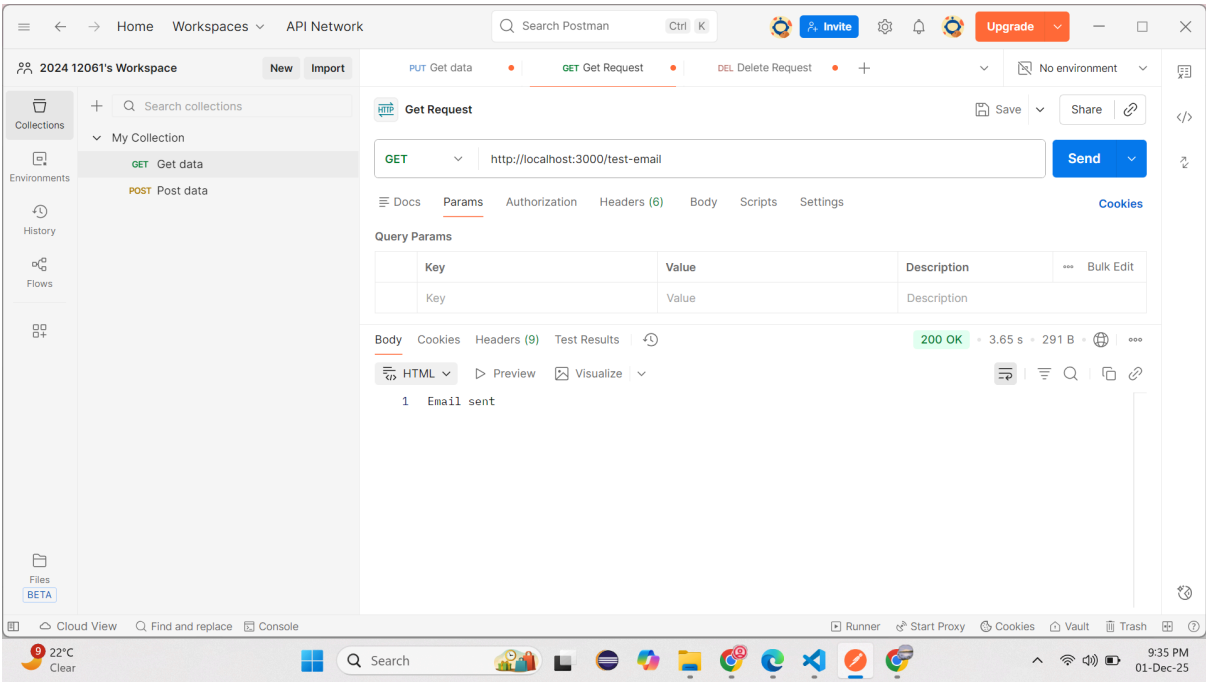
202412061



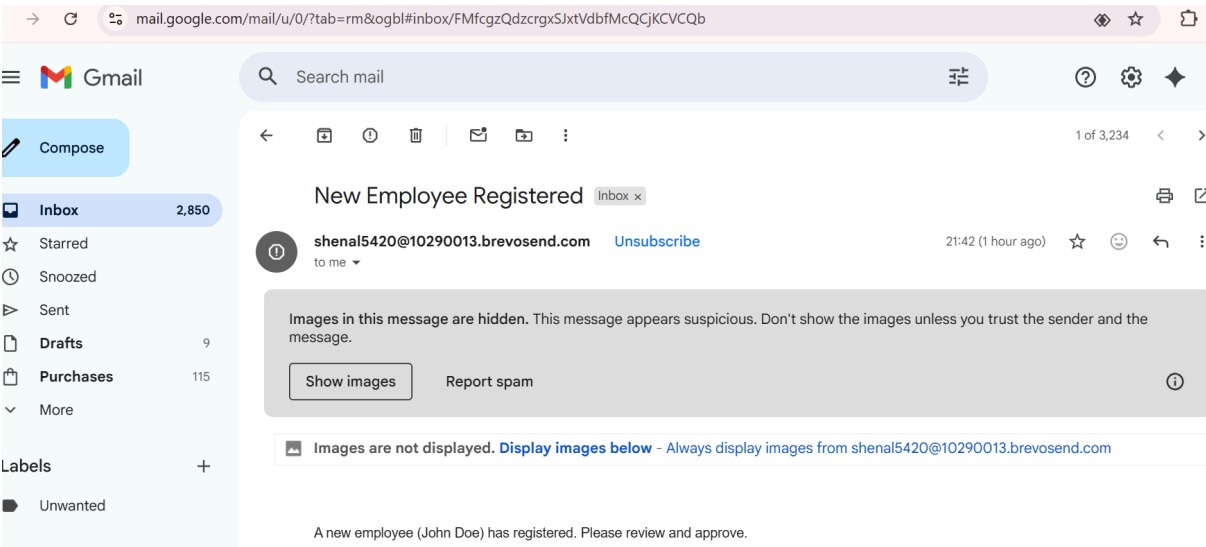
patch:



get:

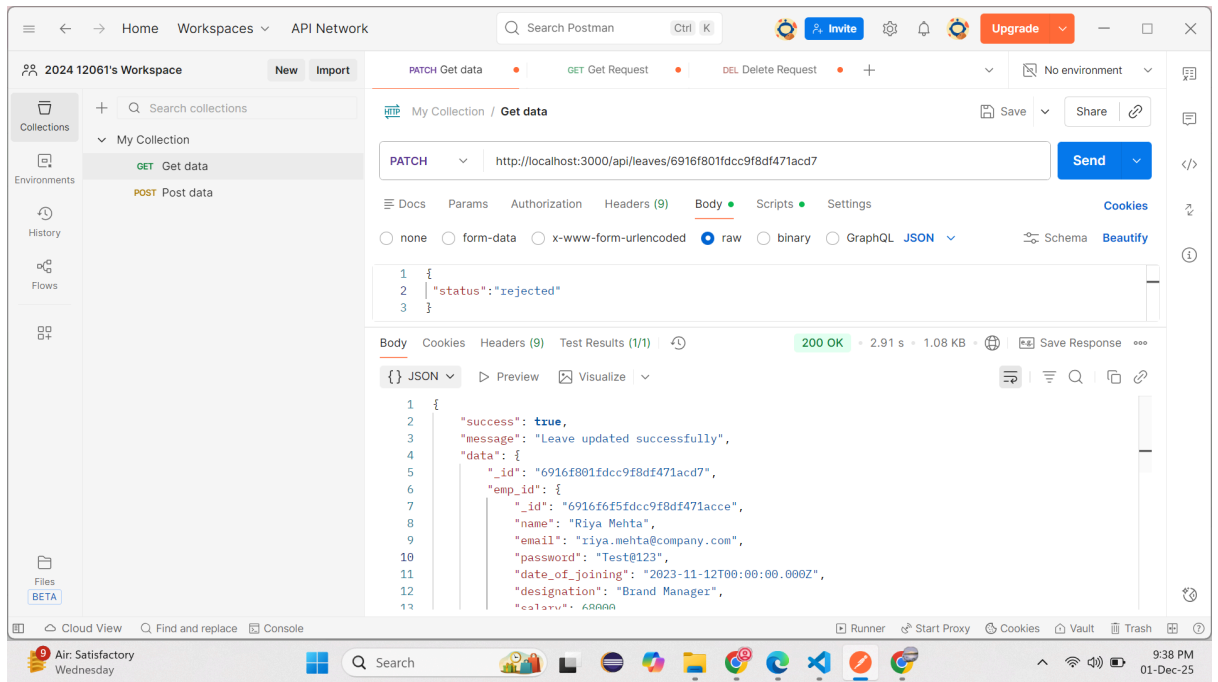


Email sample:

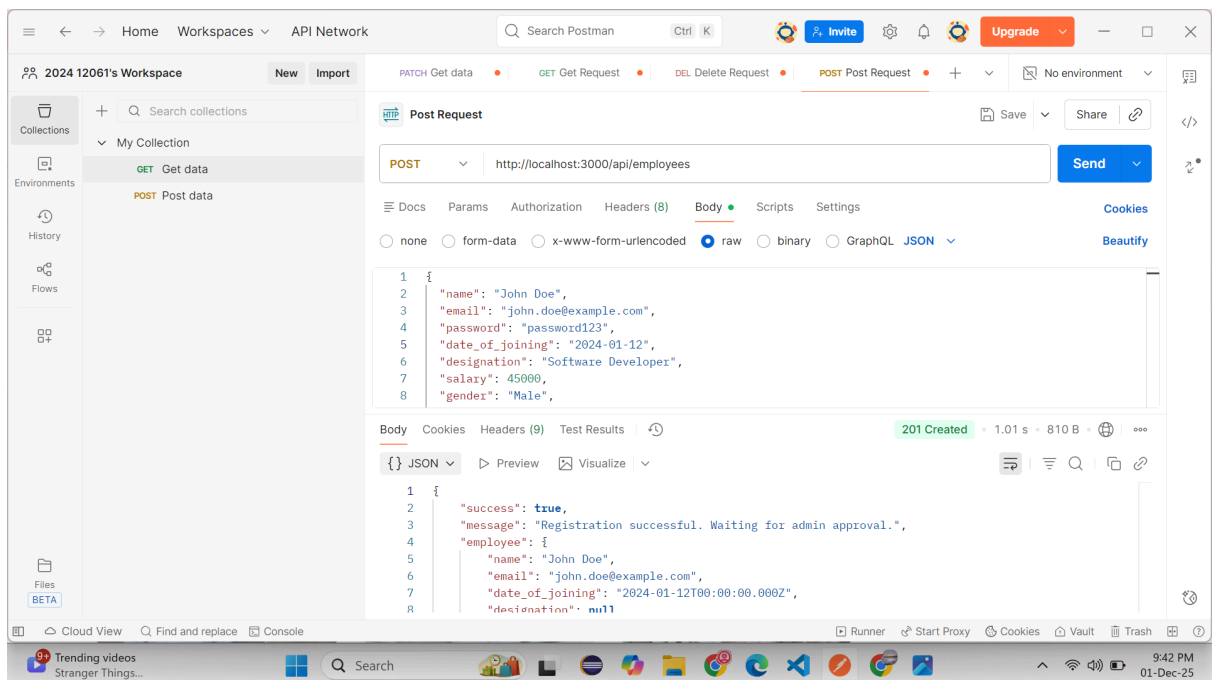


Patch:

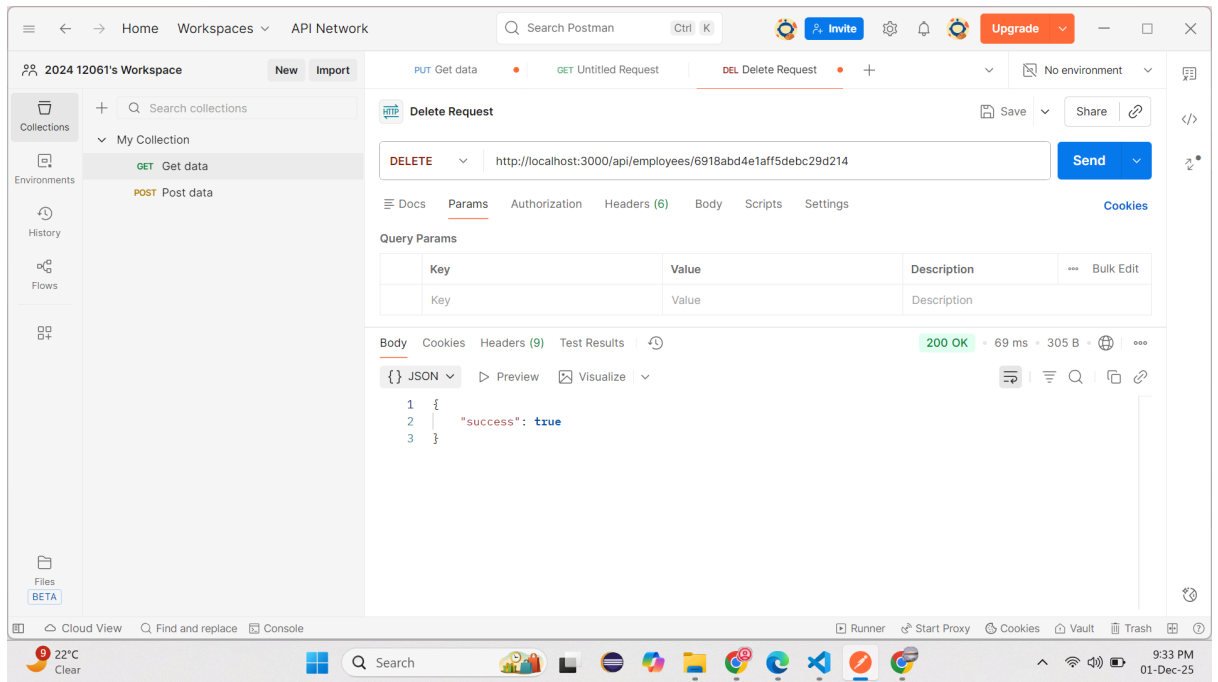
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Post:



Delete:



Acknowledgements

- MongoDB Atlas
 - Node.js & React.js communities
 - Tailwind & Bootstrap teams
 - Socket.IO & Nodemailer documentation
-